Homework 5: Let's Play LLM

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Part 1: Designing a Task for LLM and Explore the Capability of LLM

Task Description

In this scenario, there are total 20 balls, with 5 red, 5 yellow, 5 green and 5 blue. The task is to calculate the number of different combinations of balls. It also includes four different type of questions, which are related to calculate the combination in different condition. The ideal output should provide the correct calculation process and answer for each question.

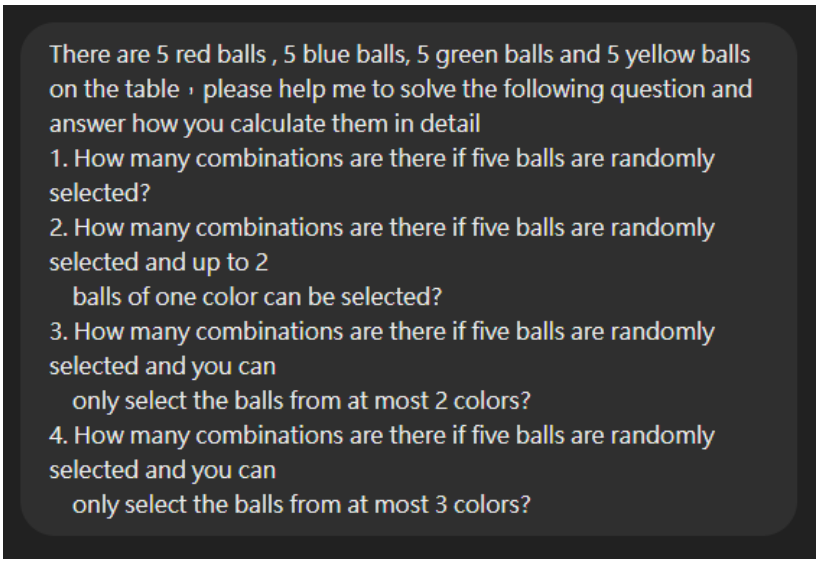
Motivation

I choose this task because it involves combinatorial mathematics calculations, which are complex with different algorithms for different situations. Through this task, I hope to explore the potential of large language models(ChatGPT) in mathematical reasoning, explanation capabilities and computational ability.

Describe your Initial Attempt

Input:

Here I only specify the total numbers of the balls and the numbers of balls of each color, and set the condition for each question.



Output of ChatGPT:

1. How many combinations are there if five balls are randomly selected?

There are a total of 20 balls (5 of each color). The number of ways to choose 5 balls out of 20 is given by the binomial coefficient:

$$\binom{20}{5} = \frac{20!}{5!(20-5)!} = \frac{20!}{5! \cdot 15!}$$

Calculating this:

$$\binom{20}{5} = \frac{20 \times 19 \times 18 \times 17 \times 16}{5 \times 4 \times 3 \times 2 \times 1} = 15504$$

So, there are 15,504 combinations.

2. How many combinations are there if five balls are randomly selected and up to 2 balls of one color can be selected?

We need to ensure that no more than 2 balls of any one color are chosen. Let's break this into cases based on the number of balls from each color:

1. 2-2-1: Two balls each from two colors and one ball from a third color.
   - Choose 2 colors out of 4 for the pairs: $\binom{4}{2}$
   - Choose 2 balls out of 5 for each of these two colors: $\binom{5}{2}$ twice.
   - Choose 1 ball out of 5 for the third color: $\binom{5}{1}$

$$\binom{4}{2} \cdot \left(\binom{5}{2}\right)^2 \cdot \binom{5}{1} = 6 \cdot 10^2 \cdot 5 = 3000$$

2. 2-1-1-1: Two balls from one color and one ball from each of the three remaining colors.
   - Choose 1 color out of 4 for the pair: $\binom{4}{1}$
   - Choose 2 balls out of 5 for this color: $\binom{5}{2}$
   - Choose 1 ball out of 5 for each of the three remaining colors: $\binom{5}{1}$ three times.

$$\binom{4}{1} \cdot \binom{5}{2} \cdot \left(\binom{5}{1}\right)^3 = 4 \cdot 10 \cdot 5^3 = 5000$$

Adding these together:

$$3000 + 5000 = 8000$$

So, there are 8,000 combinations.

3. How many combinations are there if five balls are randomly selected and you can only select the balls from at most 2 colors?

We need to consider the following cases based on the distribution of balls from two colors:

1. 5-0: All 5 balls from one color.
   - Choose 1 color out of 4: $\binom{4}{1}$
   - Choose 5 balls out of 5: $\binom{5}{5}$

$$\binom{4}{1} \cdot \binom{5}{5} = 4 \cdot 1 = 4$$

2. 4-1: Four balls from one color and one ball from another color.
   - Choose 1 color out of 4 for the 4 balls: $\binom{4}{1}$
   - Choose 4 balls out of 5 for this color: $\binom{5}{4}$
   - Choose 1 color out of the remaining 3 for the single ball: $\binom{3}{1}$
   - Choose 1 ball out of 5 for this color: $\binom{5}{1}$

$$\binom{4}{1} \cdot \binom{5}{4} \cdot \binom{3}{1} \cdot \binom{5}{1} = 4 \cdot 5 \cdot 3 \cdot 5 = 300$$

3. 3-2: Three balls from one color and two balls from another color.
   - Choose 2 colors out of 4: $\binom{4}{2}$
   - Choose 3 balls out of 5 for one color: $\binom{5}{3}$
   - Choose 2 balls out of 5 for the other color: $\binom{5}{2}$

$$\binom{4}{2} \cdot \binom{5}{3} \cdot \binom{5}{2} = 6 \cdot 10 \cdot 10 = 600$$

Adding these together:

$$4 + 300 + 600 = 904$$

So, there are 904 combinations.

4. How many combinations are there if five balls are randomly selected and you can only select the balls from at most 3 colors?

We need to consider the following cases based on the distribution of balls from three colors:

1. 5-0-0: All 5 balls from one color.
   - As calculated before: 4 combinations.
2. 4-1-0: Four balls from one color and one ball from another color.
   - As calculated before: 300 combinations.
3. 3-2-0: Three balls from one color and two balls from another color.
   - As calculated before: 600 combinations.
4. 3-1-1: Three balls from one color and one ball from each of the two other colors.
   - Choose 1 color out of 4 for the 3 balls: $\binom{4}{1}$
   - Choose 3 balls out of 5 for this color: $\binom{5}{3}$
   - Choose 2 colors out of the remaining 3 for the single balls: $\binom{3}{2}$
   - Choose 1 ball out of 5 for each of these two colors: $\binom{5}{1}$ twice.

$$\binom{4}{1} \cdot \binom{5}{3} \cdot \binom{3}{2} \cdot \left(\binom{5}{1}\right)^2 = 4 \cdot 10 \cdot 3 \cdot 5^2 = 3000$$

5. 2-2-1: Two balls each from two colors and one ball from a third color.
   - As calculated before: 3000 combinations.

Adding these together:

$$4 + 300 + 600 + 3000 + 3000 = 6904$$

So, there are 6,904 combinations.

In summary:

1. 15,504 combinations if five balls are randomly selected.
2. 8,000 combinations if up to 2 balls of one color can be selected.
3. 904 combinations if balls can only be selected from at most 2 colors.
4. 6,904 combinations if balls can only be selected from at most 3 colors.

It shows that ChatGPT has the ability to handle complex mathematical questions, but the way it calculates is not always correct which might lead to wrong answer.
However, generating wrong answers either shows the limitation of ChatGPT or the prompt is not precise enough.

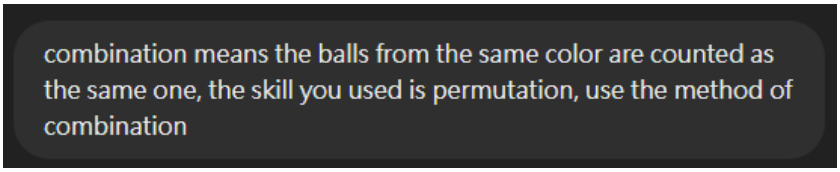
Analysis:

For question 1-4, all four questions are provided with wrong answers. By analyzing how ChatGPT calculate it, I observe that it uses the wrong method to solve the questions, it can't tell clearly that combination is different from arrangement, which the ball chosen from the same color is counted as the same one. Then, it gives the answer using the way to calculate the arrangement. Therefore, I provide the information that explain the difference between combination and arrangement, enabling ChatGPT to complete the task.

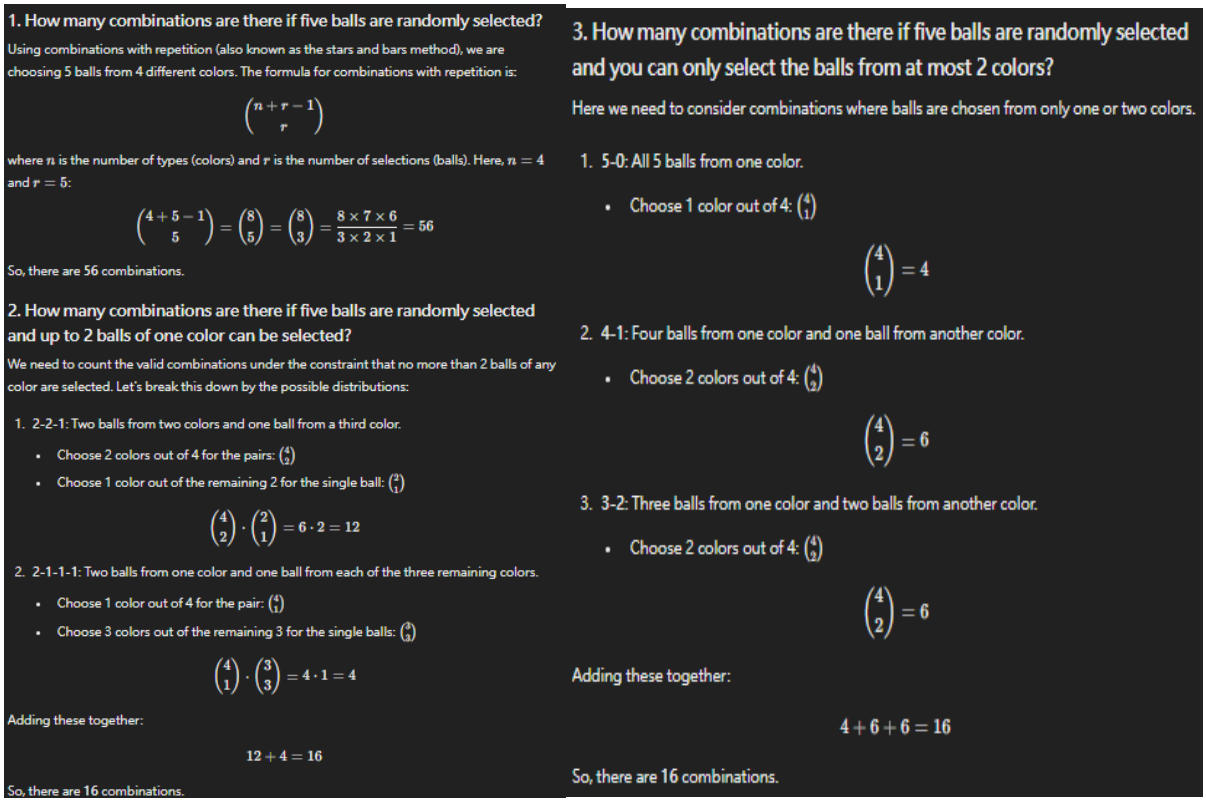
Describe Improvements 1

Input:

Here I specify the definition of combination and tell GPT to use the method of combination.



Output:



1. How many combinations are there if five balls are randomly selected?

Using combinations with repetition (also known as the stars and bars method), we are choosing 5 balls from 4 different colors. The formula for combinations with repetition is:

$$\binom{n+r-1}{r}$$

where n is the number of types (colors) and r is the number of selections (balls). Here, $n = 4$ and $r = 5$:

$$\binom{4+5-1}{5} = \binom{8}{5} = \binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

So, there are 56 combinations.

2. How many combinations are there if five balls are randomly selected and up to 2 balls of one color can be selected?

We need to count the valid combinations under the constraint that no more than 2 balls of any color are selected. Let's break this down by the possible distributions:

1. 2-2-1: Two balls from two colors and one ball from a third color.
   - Choose 2 colors out of 4 for the pairs: $\binom{4}{2}$
   - Choose 1 color out of the remaining 2 for the single ball: $\binom{2}{1}$

$$\binom{4}{2} \cdot \binom{2}{1} = 6 \cdot 2 = 12$$

2. 2-1-1-1: Two balls from one color and one ball from each of the three remaining colors.
   - Choose 1 color out of 4 for the pair: $\binom{4}{1}$
   - Choose 3 colors out of the remaining 3 for the single balls: $\binom{3}{3}$

$$\binom{4}{1} \cdot \binom{3}{3} = 4 \cdot 1 = 4$$

Adding these together:

$$12 + 4 = 16$$

So, there are 16 combinations.

3. How many combinations are there if five balls are randomly selected and you can only select the balls from at most 2 colors?

Here we need to consider combinations where balls are chosen from only one or two colors.

1. 5-0: All 5 balls from one color.
   - Choose 1 color out of 4: $\binom{4}{1}$

$$\binom{4}{1} = 4$$

2. 4-1: Four balls from one color and one ball from another color.
   - Choose 2 colors out of 4: $\binom{4}{2}$

$$\binom{4}{2} = 6$$

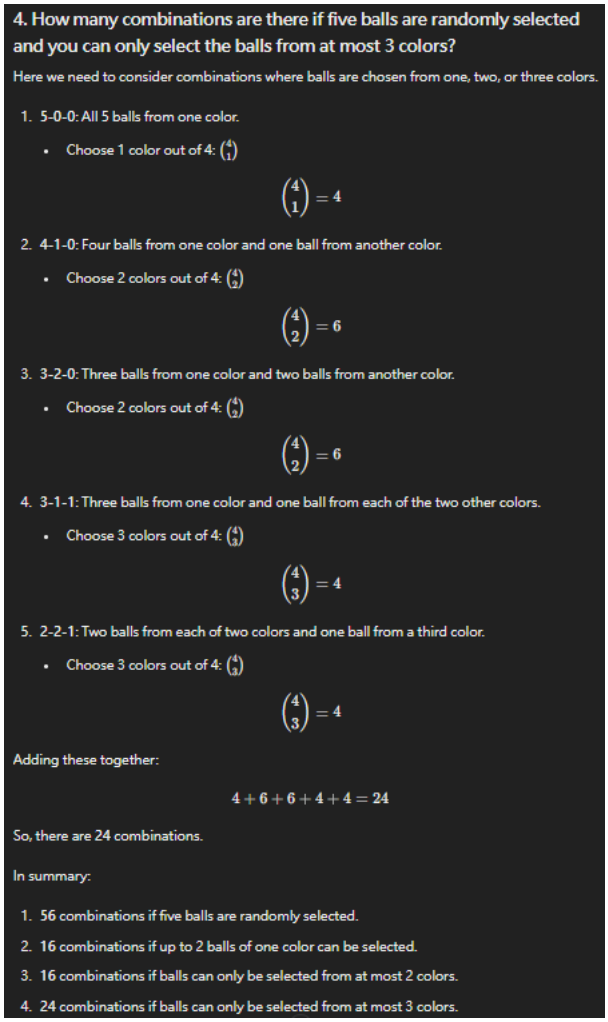
3. 3-2: Three balls from one color and two balls from another color.
   - Choose 2 colors out of 4: $\binom{4}{2}$

$$\binom{4}{2} = 6$$

Adding these together:

$$4 + 6 + 6 = 16$$

So, there are 16 combinations.



4. How many combinations are there if five balls are randomly selected and you can only select the balls from at most 3 colors?

Here we need to consider combinations where balls are chosen from one, two, or three colors.

1. 5-0-0: All 5 balls from one color.
   - Choose 1 color out of 4: $\binom{4}{1}$

$$\binom{4}{1} = 4$$

2. 4-1-0: Four balls from one color and one ball from another color.
   - Choose 2 colors out of 4: $\binom{4}{2}$

$$\binom{4}{2} = 6$$

3. 3-2-0: Three balls from one color and two balls from another color.
   - Choose 2 colors out of 4: $\binom{4}{2}$

$$\binom{4}{2} = 6$$

4. 3-1-1: Three balls from one color and one ball from each of the two other colors.
   - Choose 3 colors out of 4: $\binom{4}{3}$

$$\binom{4}{3} = 4$$

5. 2-2-1: Two balls from each of two colors and one ball from a third color.
   - Choose 3 colors out of 4: $\binom{4}{3}$

$$\binom{4}{3} = 4$$

Adding these together:

$$4 + 6 + 6 + 4 + 4 = 24$$

So, there are 24 combinations.

In summary:

1. 56 combinations if five balls are randomly selected.
2. 16 combinations if up to 2 balls of one color can be selected.
3. 16 combinations if balls can only be selected from at most 2 colors.
4. 24 combinations if balls can only be selected from at most 3 colors.

It shows that ChatGPT has the ability to employ the method of combination, but there are still some cases that it gives the wrong answer. Hence, I give a more accurate prompt to ChatGPT.

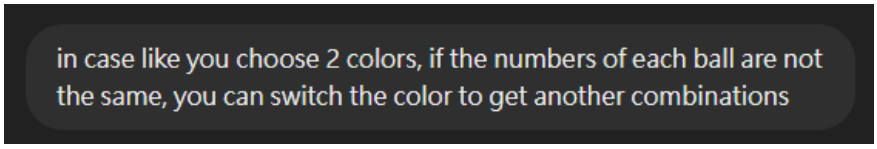
Analysis:

For question 1-2, the questions are easier because they do not restrict the color that ChatGPT can choose from. However, question 3-4 restrict the specific number of the color to choose, so it's a little bit different from question 1-2, and that means I have to give more information to ChatGPT to get the correct answer for those cases. Therefore, I provide the case which at most 2 colors can be chosen, guide the GPT to discuss the correct answer for those cases.

Describe Improvements 2

Input:

Here I specify that when choosing from 2 colors, if the numbers of each ball are not the same, you can switch the color to get another combination.



Output:

1. How many combinations are there if five balls are randomly selected?

Using combinations with repetition (also known as the stars and bars method), we are choosing 5 balls from 4 different colors. The formula for combinations with repetition is:

$$\binom{n+r-1}{r}$$

where n is the number of types (colors) and r is the number of selections (balls). Here, $n = 4$ and $r = 5$:

$$\binom{4+5-1}{5} = \binom{8}{5} = \binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

So, there are 56 combinations.

2. How many combinations are there if five balls are randomly selected and up to 2 balls of one color can be selected?

We need to count the valid combinations under the constraint that no more than 2 balls of any color are selected. Let's break this down by the possible distributions:

1. 2-2-1: Two balls each from two colors and one ball from a third color.
   - Choose 2 colors out of 4 for the pairs: $\binom{4}{2}$
   - Choose 1 color out of the remaining 2 for the single ball: $\binom{2}{1}$

$$\binom{4}{2} \cdot \binom{2}{1} = 6 \cdot 2 = 12$$

2. 2-1-1-1: Two balls from one color and one ball from each of the three remaining colors.
   - Choose 1 color out of 4 for the pair: $\binom{4}{1}$
   - Choose 3 colors out of the remaining 3 for the single balls: $\binom{3}{3}$

$$\binom{4}{1} \cdot \binom{3}{3} = 4 \cdot 1 = 4$$

Adding these together:

$$12 + 4 = 16$$

So, there are 16 combinations.

3. How many combinations are there if five balls are randomly selected and you can only select the balls from at most 2 colors?

Here we need to consider combinations where balls are chosen from only one or two colors.

1. 5-0: All 5 balls from one color.
   - Choose 1 color out of 4: $\binom{4}{1}$

$$\binom{4}{1} = 4$$

2. 4-1: Four balls from one color and one ball from another color.
   - Choose 2 colors out of 4: $\binom{4}{2}$
   - There are 2 distinct ways to distribute the 5 balls: (4,1) and (1,4).

$$\binom{4}{2} \cdot 2 = 6 \cdot 2 = 12$$

3. 3-2: Three balls from one color and two balls from another color.
   - Choose 2 colors out of 4: $\binom{4}{2}$
   - There are 2 distinct ways to distribute the 5 balls: (3,2) and (2,3).

$$\binom{4}{2} \cdot 2 = 6 \cdot 2 = 12$$

Adding these together:

$$4 + 12 + 12 = 28$$

So, there are 28 combinations.

4. How many combinations are there if five balls are randomly selected and you can only select the balls from at most 3 colors?

Here we need to consider combinations where balls are chosen from one, two, or three colors.

1. 5-0-0: All 5 balls from one color.
   - Choose 1 color out of 4: $\binom{4}{1}$

$$\binom{4}{1} = 4$$

2. 4-1-0: Four balls from one color and one ball from another color.
   - Choose 2 colors out of 4: $\binom{4}{2}$
   - There are 2 distinct ways to distribute the 5 balls: (4,1) and (1,4).

$$\binom{4}{2} \cdot 2 = 6 \cdot 2 = 12$$

3. 3-2-0: Three balls from one color and two balls from another color.
   - Choose 2 colors out of 4: $\binom{4}{2}$
   - There are 2 distinct ways to distribute the 5 balls: (3,2) and (2,3).

$$\binom{4}{2} \cdot 2 = 6 \cdot 2 = 12$$

4. 3-1-1: Three balls from one color and one ball from each of the two other colors.
   - Choose 3 colors out of 4: $\binom{4}{3}$
   - There are 6 distinct ways to distribute the 5 balls: (3,1,1), (1,3,1), (1,1,3), (3,1,1), (1,3,1), (1,1,3).

$$\binom{4}{3} \cdot 6 = 4 \cdot 6 = 24$$

5. 2-2-1: Two balls each from two colors and one ball from a third color.
   - Choose 3 colors out of 4: $\binom{4}{3}$
   - There are 3 distinct ways to distribute the 5 balls: (2,2,1), (2,1,2), (1,2,2).

$$\binom{4}{3} \cdot 3 = 4 \cdot 3 = 12$$

Adding these together:

$$4 + 12 + 12 + 24 + 12 = 64$$

So, there are 64 combinations.

In summary:

1. 56 combinations if five balls are randomly selected.
2. 16 combinations if up to 2 balls of one color can be selected.
3. 28 combinations if balls can only be selected from at most 2 colors.
4. 64 combinations if balls can only be selected from at most 3 colors.

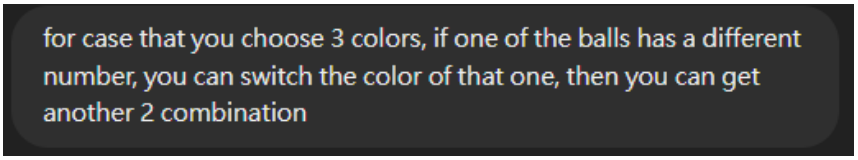
Analysis:

There is still one mistake which is in question that 3 colors can be chosen, and ChatGPT count the 3-1-1 case twice, so I specify that for case choosing from 3 colors, if one of the balls has a different number, you can switch the color of that one, then you can get another 2 combinations.

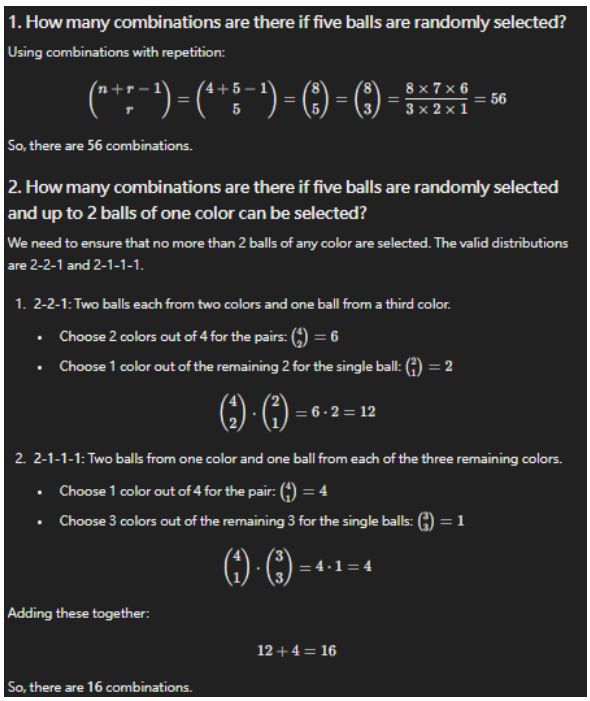
Describe Improvements 3

Input:

Here I specify that when choosing from 3 colors, if one of the balls has a different number, you can switch the color of that one, then you can get another 2 combinations.



Output:



1. How many combinations are there if five balls are randomly selected?

Using combinations with repetition:

$$\binom{n+r-1}{r} = \binom{4+5-1}{5} = \binom{8}{5} = \binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

So, there are 56 combinations.

2. How many combinations are there if five balls are randomly selected and up to 2 balls of one color can be selected?

We need to ensure that no more than 2 balls of any color are selected. The valid distributions are 2-2-1 and 2-1-1-1.

1. 2-2-1: Two balls each from two colors and one ball from a third color.
   - Choose 2 colors out of 4 for the pairs: $\binom{4}{2} = 6$
   - Choose 1 color out of the remaining 2 for the single ball: $\binom{2}{1} = 2$

$$\binom{4}{2} \cdot \binom{2}{1} = 6 \cdot 2 = 12$$

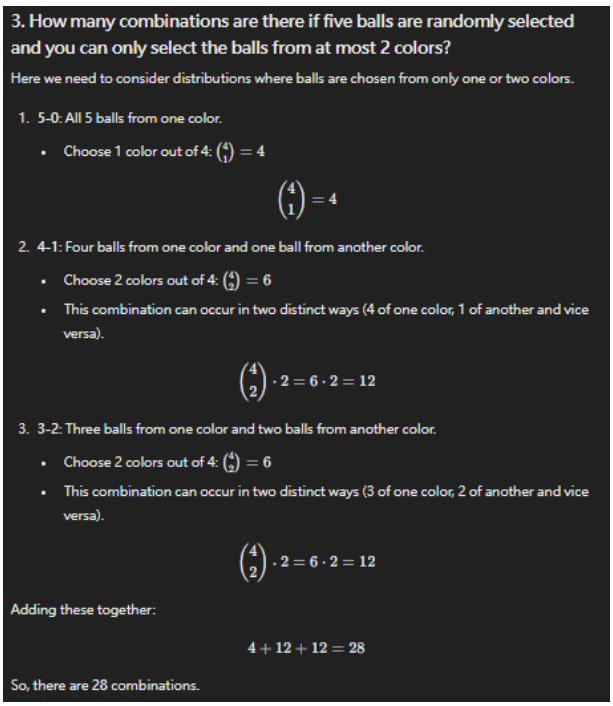
2. 2-1-1-1: Two balls from one color and one ball from each of the three remaining colors.
   - Choose 1 color out of 4 for the pair: $\binom{4}{1} = 4$
   - Choose 3 colors out of the remaining 3 for the single balls: $\binom{3}{3} = 1$

$$\binom{4}{1} \cdot \binom{3}{3} = 4 \cdot 1 = 4$$

Adding these together:

$$12 + 4 = 16$$

So, there are 16 combinations.



3. How many combinations are there if five balls are randomly selected and you can only select the balls from at most 2 colors?

Here we need to consider distributions where balls are chosen from only one or two colors.

1. 5-0: All 5 balls from one color.
   - Choose 1 color out of 4: $\binom{4}{1} = 4$

$$\binom{4}{1} = 4$$

2. 4-1: Four balls from one color and one ball from another color.
   - Choose 2 colors out of 4: $\binom{4}{2} = 6$
   - This combination can occur in two distinct ways (4 of one color, 1 of another and vice versa).

$$\binom{4}{2} \cdot 2 = 6 \cdot 2 = 12$$

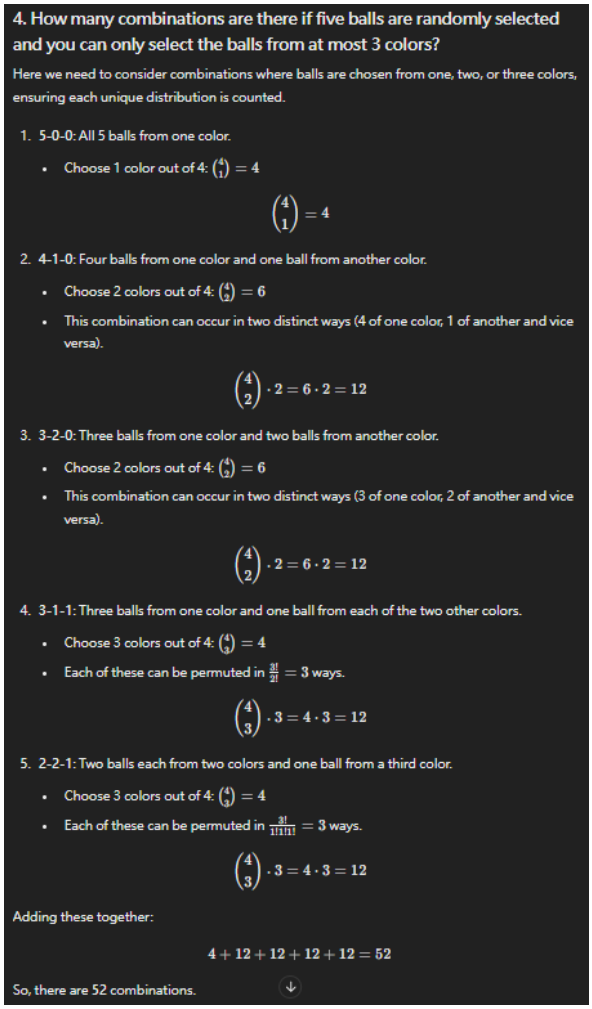
3. 3-2: Three balls from one color and two balls from another color.
   - Choose 2 colors out of 4: $\binom{4}{2} = 6$
   - This combination can occur in two distinct ways (3 of one color, 2 of another and vice versa).

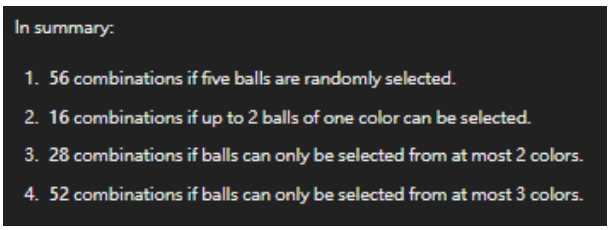
$$\binom{4}{2} \cdot 2 = 6 \cdot 2 = 12$$

Adding these together:

$$4 + 12 + 12 = 28$$

So, there are 28 combinations.





Analysis:

After these prompts, ChatGPT outputs the correct answers for question 1-4 in detail based on the prompts I give.

Part 2: Problem Answering

Question 1

1. Besides providing a more detailed description of the question in the prompt, what other methods can be employed to improve the accuracy of the responses?

1. Using Examples: Providing examples of correct and incorrect responses can help guide the model towards the desired output.
2. Clarifying Ambiguities: Explicitly stating any potential ambiguities in the prompt and clarifying them can reduce misunderstandings.
3. Defining Context and Constraints: Clearly defining the context in which the question is being asked and any constraints that must be adhered to can improve response accuracy.
4. Iterative Refinement: Refining the prompt iteratively based on the responses obtained can help in fine-tuning the instructions to get better results.
5. Specific Question Types: Asking specific types of questions such as multiple-choice or true/false can lead to more precise answers.

2. Please provide an example demonstrating the impact of implementing the aforementioned methods on a prompt.

Example Demonstrating the Impact of **Using Examples**

- Original Prompt: "Write an email to request a meeting."

Response to Original Prompt: "Hi, can we have a meeting? Thanks."

- Improved Prompt with Examples: "Write an email to request a meeting. For example, you can start with a polite greeting, explain the purpose of the meeting, suggest a few available dates and times, and end with a thank you and your contact information."

Response to Improved Prompt:
"Subject: Meeting Request to Discuss Project Updates
Dear [Recipient's Name],
I hope this email finds you well.
I am writing to request a meeting to discuss the recent updates on our project. I believe it would be beneficial for us to go over the progress and address any questions or concerns you might have.
Could you please let me know your availability for a meeting next week? Here are a few time slots that work for me:
Monday, June 14th at 10:00 AM
Tuesday, June 15th at 2:00 PM
Wednesday, June 16th at 11:00 AM
Please let me know if any of these times work for you, or suggest a different time that is more convenient.
Thank you very much for your time and consideration. I look forward to your response.
Best regards, [Your Name] [Your Contact Information]"

- Impact of Improved Prompt:

By providing an example and guiding the structure of the response, the accuracy and completeness of the LLM's response can be significantly improved.

- Analysis:

In the original prompt, the response was vague and lacked essential details that would make the email effective and professional. By using the method of providing examples, the improved prompt guided the LLM to produce a detailed and structured email that includes a clear purpose, suggested meeting times, and a polite closing. This method effectively enhances the quality and accuracy of the responses, making them more useful in real-world scenarios.

Question 2

While using LLM, you probably find that it cannot accurately answer your queries. This indicates that current large language models (LLMs) still face numerous challenges. Please search online (papers, blogs, etc.) for the existing challenges of LLMs and briefly explain the motivation, difficulties, and current methods available to optimize or solve these issues.

1. Understanding Context and Ambiguities

- Motivation: LLMs often misunderstand context or provide ambiguous responses due to limited understanding of nuanced language.
- Challenges: Accurately interpreting complex contexts, handling ambiguous queries, and maintaining coherence over long conversations.
- Current Methods:
  1. Context-Aware Training: Incorporating more contextual information during training to help the model understand and retain context better.
  2. Clarification Prompts: Designing prompts that encourage the model to ask clarifying questions when faced with ambiguous queries.

2. Reducing Bias and Ethical Concerns

- Motivation: Bias in LLMs can lead to harmful or unfair outputs, impacting user trust and ethical use.
- Challenges: Identifying and mitigating various biases present in the training data, ensuring fair and unbiased responses across different demographics.
- Current Methods:
  1. Bias Detection Algorithms: Implementing algorithms to detect and mitigate biases in LLM responses.
  2. Diverse Training Data: Using diverse and balanced datasets to train LLMs, reducing the risk of biased outputs.